Conditional generation of cathode vacancy patterns: separating latent learning, spatial decoding, and relaxed energy

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Abstract

We test whether a conditional variational autoencoder (CVAE) proposes new Li/vacancy configurations at a requested composition and energy range more efficiently than uniform sampling and mutation. Earlier spatial decoding improved teacher-defined yield, but its latent variables were nearly unused. The present matched controls separate a graph-posterior CVAE, a no-latent autoregressive sampler, and a dense-posterior CVAE with the same causal spatial decoder. Thirty-seven 4,000-update fits are complete. Development evidence shows spatial AR matches the graph CVAE, while a dense posterior retains information without useful prior-generation gain. On frozen FINAL proposals, radial-shell AR achieves 63.04% UMA balanced novel in-bin yield, versus 62.20% for its CVAE and 58.37% for mutation. A separate 6,000-mask cohort compares deletion and relaxed-endpoint targets; endpoint-label training does not improve endpoint LOW yield in the matched test. Exhaustive small-cell enumeration quantifies approximate symmetry and teacher disagreement. No new DFT or interatomic-potential fine-tuning is performed, and generative, numerical, and physical validation remain distinct.

1 Motivation, revised question, and independent data

Direct recycling aims to restore useful cathode composition and structure rather than dissolve the material into its constituent elements. Hydrothermal processing and annealing have regenerated spent $LiCoO_2$ (LCO) experimentally [1]. Our longer-term motivation is to select informative atomistic configurations for reference calculations and improved recycling models. The class project studies conditional generation of vacancy patterns; it does not simulate a hydrothermal liquid or establish optimal reactor conditions.

Density functional theory (DFT) supplies an approximate electronic-structure reference at specified atoms, positions, and cell. A machine-learned interatomic potential (MLIP) predicts potential energy and its force/cell derivatives. A single-point prediction leaves geometry unchanged. Relaxation moves atoms toward a local minimum; molecular dynamics (MD) uses forces to propagate them in time. A sampled vacancy mask performs none of these operations automatically.

Li/vacancy ordering changes energy even at fixed composition [2]. The working parent is $Li_{48}Co_{48}O_{96}$. An occupation vector $o \in \{0,1\}^{48}$ has $o_j = 1$ for a present Li at parent site j and $o_j = 0$ for a vacancy. Its count is $n = \sum_j o_j$; the material contains $N_a = 144 + n$ atoms and Li fraction $x = n/48$. Primary counts are $n = 12, 24, 36$. The parent geometry and cell are fixed for deletion labels. The primary teacher is UMA-S-1.2.1/`omat`; the independent teacher check is MACE-MH-1/`omat_pbe`. Both use FP32, TF32 off, with checkpoint and inference receipts. A missing precision field in an old condition schema is not evidence that its predictions used another precision.

The parent's 192 retained site permutations define an *orbit*: all vacancy masks related by those operations. The lexicographically smallest transformed vacancy string identifies the orbit. Grouping precedes splitting; symmetry augmentation supplies equivalent views, not additional independent examples. The fixed round-0 dataset contains 40,157 TRAIN, 4,967 VAL and 5,123 TEST orbits. Only TRAIN supplies fit examples and class thresholds; VAL selects checkpoints. TEST labels are not used for these decisions. Before FINAL generation, a history of 344,001 known orbits was frozen, including development proposals, acquired structures and observed research trajectories. Novelty uses this complete history.

The revised scientific question is whether an improvement is due to the CVAE latent pathway, to causal spatial autoregression, or to their combination, and whether a deletion-energy advantage survives actual geometry relaxation. This is more discriminating than comparing a CVAE only against uniform masks.

2 Model, conditions, and controlled alternatives

The condition is $c = (n/48, b)$, with energy class $b \in \{\text{low}, \text{middle}, \text{high}\}$. Within each count, TRAIN energy quantiles q_{20}, q_{80} define the frozen class boundaries. Energy is in eV per present atom for class assignment; the teacher, parent, geometry convention and thresholds belong to one condition schema. A low request is a supported energy *range*, not a guarantee of a global minimum or an energy-gradient optimization.

An encoder produces a diagonal Gaussian $q_\phi(z \mid o, c) = \mathcal{N}(\mu_\phi, \operatorname{diag}\sigma_\phi^2)$, where $z \in \mathbb{R}^4$ is the latent vector. The reparameterization $z = \mu + \sigma \odot \epsilon$, $\epsilon \sim \mathcal{N}(0, I)$, permits stochastic training. At generation, a fresh $z \sim \mathcal{N}(0, I)$ is used without a target mask. The conditional autoregressive decoder factors

$$p_\theta(o \mid z, c) = \prod_{i=1}^{48} p_\theta(o_i \mid o_{<i}, z, c, h_i). \tag{1}$$

Here $o_{<i}$ is the already assigned prefix, and h_i summarizes local occupied, vacant and unassigned sites. At training, the true prefix is supplied (teacher forcing); at generation, earlier sampled decisions form the prefix. The future target occupations are excluded from h_i. If the remaining required Li count is zero, later sites are vacant; if it equals the remaining site count, all are occupied. This enforces $\sum o_i = n$ exactly.

For one training row, the loss is

$$\mathcal{L}(o, c) = -\mathbb{E}_{q_\phi} \sum_i \log p_\theta(o_i \mid o_{<i}, z, c, h_i) + \beta_t D_{\mathrm{KL}}(q_\phi(z \mid o, c) \| \mathcal{N}(0, I)) \,. \tag{2}$$

The first term rewards mask reconstruction; the KL term regularizes the posterior toward the sampling prior. Both are measured in nats with natural logarithms. The same row weights multiply reconstruction and the sum of KL over latent components. Class balancing changes TRAIN weights only; VAL retains its base weights and fixed Monte Carlo noise. The block-update KL warm-up and selection objective with $\beta = 0.1$ are matched across controls. All full fits use 4,000 updates: 37 trajectories total, including bounded screens and six paired-target fits, for 148,000 updates.

Control	Posterior / latent	Shared decoder and discriminating question
A	Graph encoder, $z \in \mathbb{R}^4$	Physical causal neighborhoods. Does the spatial CVAE use a latent representation?
B	No encoder or latent	Same causal neighborhoods. Is a conditional spatial AR sampler sufficient?
C	Dense encoder, $z \in \mathbb{R}^4$	Same causal neighborhoods. Does an informative dense posterior improve prior sampling?
D	No latent, matched B	Fixed nonphysical association of decoder graph kernels. Full kernel degree/weights are retained, but causal prefix exposure also changes.

The graph encoder passes messages on the full parent, including host species and vacant mobile sites, then pools host/mobile features [3]. The causal decoder uses only prefix-derived features on that fixed graph. Its six-feature baseline is compared with nine occupied/vacant/unassigned fractions in three radial shells, including interlayer neighbors. Three-fit development confirmation retained A-shell as the best genuine CVAE point estimate and B-shell as the strongest AR; A-shell's improvement over A was unresolved across fits. A/B and their shell variants have five FINAL replicates; C/D/remaining-count B have three. Replicates 0804/05 were FINAL-only. This is not a coordinate-generating equivariant model. The software accepts a different parent/mobile ion, but a new material or cell size has not been validated.

KL, per-component $\mathrm{Var}(\mu_j)$, resampling z with shared decoder uniforms, and changing b with shared z/uniforms test different properties. A small reconstruction loss alone does not establish useful latent codes [4]. A cyclic KL policy is a future controlled candidate [5], not an assumed cure for collapse or evidence of better sampling.

3 Evaluation and mechanistic evidence

Uniform masks select exactly n sites. Mutation starts from a frozen TRAIN seed in the requested bin and swaps equal numbers of present/vacant sites. Seed information, proposal counts and swap budgets are declared. Neither baseline is credited with a new unique structure when it merely repeats one already seen. Every attempted proposal remains in the denominator, including duplicates, unsupported/failed requests and invalid outputs.

For A_{snb} attempted proposals from replicate s, count n and requested bin b, the primary score is

$$Y_{snb} = \frac{\#\{\text{distinct valid orbits, new to frozen history, teacher bin} = b\}}{A_{snb}}. \tag{3}$$

The balanced yield averages all nine count/bin cells equally and is then reported per fitted replicate. The historical low-energy score using its original round-0 q_{20} remains a separately named metric. Best energy versus attempts and versus actual new teacher queries are secondary search metrics; they need not agree with requested-bin yield. Shared queries are deduplicated physically without changing attempted-proposal denominators.

Paired condition-response tests hold z and decoder uniforms fixed while changing the requested energy class; independent MLIP rescoring determines the realized energy shift. Latent-response tests change z while fixing condition/uniforms. Paired probes report raw Hamming differences and orbit equality, failures, novelty, energy distributions and composition-resolved results. FINAL structural distance minimizes over all 192 operations against 1,024 fixed TRAIN orbits/count; it is an upper bound on full-TRAIN nearest distance. A few fit seeds give descriptive variation, not a calibrated confidence interval over all future fits.

Method	Fits	Balanced yield (%, mean±SD) UMA	MACE	Original-q_{20} LOW (%) UMA	MACE
A: graph CVAE	5	60.24 ± 0.53	61.72 ± 0.72	62.81	66.25
B: spatial AR	5	60.21 ± 1.23	62.03 ± 0.76	62.24	66.61
C: dense CVAE	3	31.37 ± 0.22	31.02 ± 0.96	32.47	36.89
D: nonphysical AR	3	34.38 ± 0.80	35.04 ± 0.78	25.43	28.82
A-shell: graph CVAE	5	62.20 ± 0.56	64.03 ± 0.79	63.33	67.66
B-shell: spatial AR	5	63.04 ± 1.47	64.38 ± 0.96	65.62	69.43
B: remaining count	3	61.40 ± 0.94	63.28 ± 0.31	63.02	67.45
Mutation	5	58.37 ± 1.08	57.97 ± 1.39	50.89	51.93
Uniform	5	33.32 ± 0.04	33.32 ± 0.04	20.99	21.04

These are frozen **FINAL** attempts, with descriptive fit/sampler SD in percentage points. C/D paired controls use the first three matching A/B fits; individual fits and paired contrasts are retained in the packet. Uniform is 33.316%, not exactly one third: small repeat/novelty losses remain. D changes physical association and prefix exposure, so it is not a pure locality ablation.

The initial mechanism check found A's VAL KL at 0.006–0.019 nats/row with no active components at $\mathrm{Var}(\mu_j) > 0.01$, versus C's 12.5–13.3 nats and four active components. On FINAL, paired LOW→HIGH shifts UMA energy by 5.37/5.36 meV/atom for A/B, but only 0.33 for C; shell A/B shifts are 5.97/5.94. Resampling z changes shell A by 0.851 sites on average (65.42% same orbit), versus C's 13.04 sites (0.26%). Mean absolute energy changes are 0.335 versus 2.186 meV/atom. Signed mean change near zero is expected for exchangeable prior draws and is not collapse evidence. Low KL and weak mask response support weak latent use; an active posterior does not establish useful prior generation. Across five fits, A-shell minus B-shell balanced yield is -0.83 ± 1.13 UMA and -0.35 ± 0.66 MACE percentage points (descriptive SD), without a demonstrated variational advantage.

The FINAL static manifest freezes 44,928 primary attempts and 4,992 separate LOW latent probes across 39 model/baseline entries. MACE scores the exact same geometries under its own frozen schema. Uniform uses shared draws across requested classes: without repeat/novelty losses, exactly one-third balanced yield follows algebraically from this coupling, not zero uncertainty in individual class yields. The FINAL loss and nonzero fit spread are retained. Novelty uses the complete frozen history. FINAL outcomes do not select or modify models. Teacher agreement is robustness evidence, not calibrated uncertainty or DFT validation.

4 Does deletion-energy control survive relaxation?

For the same initial mask o, define $E_{\mathrm{del}}(o)$ at literal parent coordinates and $E_{\mathrm{end}}(o)$ after fixed-cell atomic relaxation. The latter is a reached local endpoint, not a certified global minimum. Two models have identical input masks, splits, architecture, initialization policy and update budget, but distinct TRAIN-fitted condition thresholds for these two energy targets. Both generators must be evaluated under *both* energy definitions. A mask generator conditioned on endpoint energy still outputs a pristine-derived deletion geometry until relaxation is actually run.

The frozen 6,000-mask cohort has 1,200 TRAIN, 400 VAL and 400 TEST masks/count. UMA fixed-cell FIRE uses maximum force 0.05 eV/Å, at most 500 steps and a 120-s attempt cap; coordinates and atom identities are retained. Of 6,000 attempts, 5,992 converged and eight were interrupted. The shared fit set has 3,594 TRAIN and 1,200 VAL pairs. Per-count TRAIN LOW support is 240; endpoint VAL LOW support is 77/81/85. This passes the predeclared minimum support of 30/count/role, an operational heuristic rather than a power calculation. TEST was sealed until the final scoring gate; 1,198/1,200 TEST pairs are known, with both failures retained and no threshold refitting.

At $n = 12/24/36$, independent TEST deletion–endpoint Spearman rank correlation is 0.926/0.966/0.907. Of deletion-LOW masks, 74.7/82.6/77.6% also reach endpoint LOW; they recover 73.8/85.1/78.6% of endpoint-LOW masks. Thus deletion energy is a useful but imperfect proxy. Mean relaxation lowers energy by 30.52/20.84/11.91 meV/atom. A high correlation alone cannot establish a generator's endpoint low-tail enrichment.

Generated arm	Deletion assessment (%)	Endpoint assessment (%)
Current LOW requests only; paired-calibration LOW thresholds		
A: current graph CVAE	64.76	65.63
B: current spatial AR	63.54	64.06
Current mutation	57.12	53.82
Uniform	21.53	22.92
Balanced three-class requests; matched new-cohort training		
Deletion-target CVAE	61.28	59.14
Endpoint-target CVAE	59.03	58.74
Deletion-seed mutation	57.29	54.28
Endpoint-seed mutation	55.96	58.56
Shared uniform	33.33	33.33

Means use three replicates and all attempted proposals. The two panels have different request distributions and training cohorts; their numbers are not direct architecture contrasts. All 7,616 physical endpoint queries converged. Endpoint-target training does not improve balanced endpoint yield over matched deletion-target training in this assessment; its endpoint-seeded mutation control is nearly tied. On endpoint LOW requests, deletion-trained versus endpoint-trained CVAE yields are 62.50 versus 60.42%; the paired difference is -2.08 ± 2.76 percentage points (descriptive SD), positive in 0/3 fits. Current A minus B's endpoint LOW difference is $+1.56 \pm 2.76$ points, positive in 2/3 fits; a latent-specific gain remains unresolved. Using the separately named original H0 q_{20} on deletion energies, current A/B/mutation/uniform LOW yields are 61.81/60.94/53.65/19.27%; that threshold is never applied to endpoint energies.

The paired FINAL manifest retains 10,944 attempts: 2,304 current LOW-only attempts (A, B, mutation, uniform), 3,456 per matched deletion/endpoint CVAE arm, and 1,728 shared uniform attempts. Each target arm is assessed under both schemas; the same 576 uniform LOW draws are shared, not independent repeats. Unknown labels and nonconvergence reduce all-attempt yield rather than disappearing. Occupation-key coincidence is not evidence of a common physical PES basin.

Tightening 96 TRAIN endpoints to 0.01 eV/Å gives 1/4/0 class crossings among 32/count. On the same 96 UMA geometries, UMA–MACE rank correlations are 0.977/0.941/0.769. MACE median maximum forces are 0.212/0.261/0.302 eV/Å, despite UMA values near 0.05: a UMA endpoint is not generally stationary for MACE. Independent MACE quenches start from deletion geometries.

5 Interpretation, limits, and independent completion tasks

The principal mechanistic distinction is between *condition control* and *latent use*. A decoder may respond well to a requested energy bin while effectively ignoring z. Conversely, an encoder can reconstruct a training mask through an informative posterior yet generate poorly from its prior. The present A/B/C controls directly separate these possibilities. Retaining a true CVAE is useful only if its measured behavior supports the latent pathway; an AR result that matches or wins remains scientifically informative. At this data/update budget, B-shell has the best balanced point estimate under both teachers, while a latent-specific advantage is unestablished. The current learned samplers enrich the endpoint LOW tail, but matched endpoint-label training does not improve it over deletion-label training. These observations support spatial conditional sampling without claiming that CVAEs or endpoint labels cannot help under another controlled design.

Pristine-derived deletions omit relaxation, local strain, magnetic degrees of freedom and competing host structures. Endpoint labels address one of these limits under a stated MLIP and fixed cell, without proving a ground state or validated chemistry. Class conditions are empirical ranges, not thermodynamic populations. The exact small-cell probabilities provide a finite-model reference for raw reference-pool error, not a justification for treating a generated proposal histogram as an equilibrium distribution.

The separate research supplement reports completed bounded checks. Exhaustive 12-site enumeration scores all 1,364 raw masks plus repeat controls under both teachers. Approximate symmetry has measurable energy spread, so every raw mask is retained. Finite-model configurational entropies differ between teachers; they are not experimental LCO thermodynamics or evidence of network size transfer. A null test using the actual 537-mode harmonic spectrum exposes finite-window bias and spurious fluidicity in thermal estimators, not an MD measurement of anharmonicity.

For one saved migration event, fixed-endpoint image/tolerance checks pass a restricted 0.02-eV path-span target, but changing endpoint relaxation changes the span by 0.0264 eV. Two finite-difference steps find one stable negative internal mode at a candidate saddle. Its downhill branches do not establish the full marked endpoint-to-endpoint hop; one reaches an off-site low-force state with the same assigned vacancy mask. Another route fails its step cap. Consequently, no stable DFT-validated hop barrier, finite-temperature activation free energy or diffusivity is claimed. Exact-coordinate viewing files and atom mappings support later review. These extensions do not substitute for the class-model comparisons.

Beihao: independently rerun and interpret the matched spatial no-latent control, verify decoder causality/count enforcement, and compare condition response with latent response. Automated fits do not constitute his independent contribution. Explain why informative C latents can coexist with poor prior sampling; keep any later ablation independent of this final assessment.

Varad: independently replay every raw-attempt denominator, frozen-history novelty, per-seed contrast, teacher-query deduplication and paired endpoint lineage/failure account. Preserve negative and null latent/target-transfer findings and distinguish proposal attempts, unique queries, actual force calls and allocated resources. Do not tune from final outcomes.

Ricardo: review the actual structures, site/host changes, force thresholds, teacher disagreement and physical scope; select later DFT checks only after the no-new-DFT restriction ends. Integrate the report and verify/manual-submit against the actual November 9, 11:59 p.m. Pacific assignment instructions. The repository supplies portable commands, cached-data replay and evidence. No additional DFT, MD or MLIP fine-tuning is needed before this midterm. The final report is due December 12, with presentations November 24 and December 1. Automated evidence is not the collaborators' independent contribution. Publication and resource closure are reported separately.

References

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